# Python Programming Foundations

Week 7 - RBA & Project Framing

*Day 2: "Constraints, Structure, & Scope Control"*

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## Smaller And Clearer Is Stronger

A smaller clear project usually beats a larger vague project.

The capstone is not judged by how many features you imagine.

It is judged by whether the project can run, be tested, and be explained.

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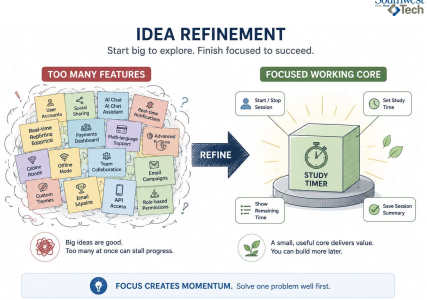


## Project Refinement

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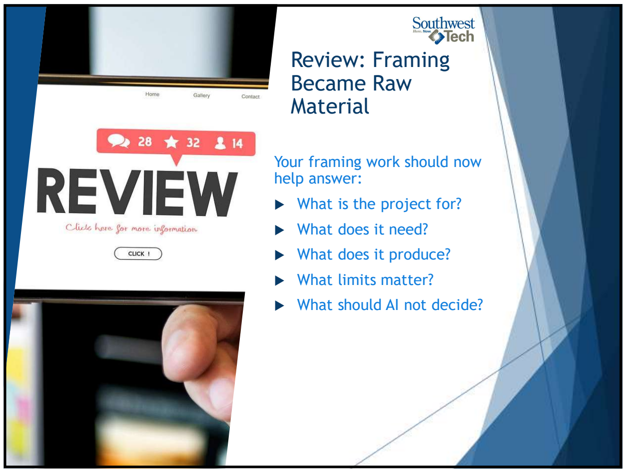
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Review: Framing Became Raw Material
REVIEW
Your framing work should now help answer:
- What is the project for?
- What does it need?
- What does it produce?
- What limits matter?
- What should AI not decide?

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FROM FRAMING TO PROPOSAL
Yesterday's clarity becomes tomorrow's plan.
A13 FRAMING
Clarify the idea
PURPOSE
What is this for?
INPUTS & OUTPUTS
What goes in and what comes out?
CONSTRAINTS & SCOPE
What are the limits and boundaries?
LIKELY STRUCTURE
What kind of solution makes sense?
RISKS
What could get in the way?
AI BOUNDARIES
What AI should and should not decide?
A clear frame creates focus.
Your answers become the foundation of your proposal.
A14 PROPOSAL
Shape the plan
PROJECT GOAL & VALUE
What will this achieve and for whom?
REQUIREMENTS
Detailed inputs and expected outputs.
SCOPE & CONSTRAINTS
What is in scope, out of scope, and limited?
SOLUTION APPROACH
The structure and key components you will build.
RISKS & MITIGATIONS
Key risks and how you will address them.
AI USE PLAN
How AI will support the work—and what stays human.
A focused plan drives results.
Good framing makes proposal writing faster, easier, and stronger.
You already did the hard part!
Review: Framing Became Raw Material

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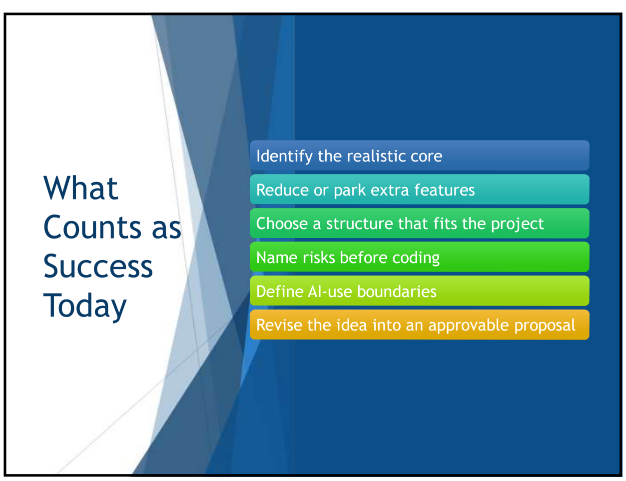
What Counts as Success Today
Identify the realistic core
Reduce or park extra features
Choose a structure that fits the project
Name risks before coding
Define AI-use boundaries
Revise the idea into an approvable proposal

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## Today's Success

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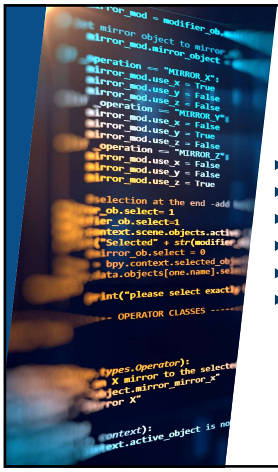

## What We Will Use Today

- constraints
- structure choices
- feature reduction
- risks and unknowns
- AI-use boundaries
- approval criteria

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## Toolbox

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## What We Will “Skip” For Now

- building all features
- polishing output
- adding advanced packages
- packaging for distribution
- asking AI to keep expanding the project

*Today we make the proposal approvable.*

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## Parked For Later

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## Project Scope

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## Constraints Are Part Of The Design

*“Good constraints make the project stronger by making reality visible.”*

- time
- skill level
- data access
- explainability
- testing difficulty
- course fit

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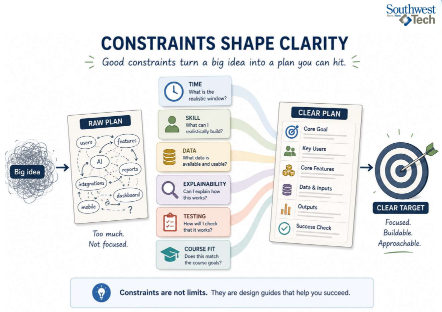


## Constraints Are Part Of The Design

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## Feature Count Is Not Project Quality

- More features can make a project weaker if they make it harder to finish or explain.
- Capstone quality comes from a coherent working core.

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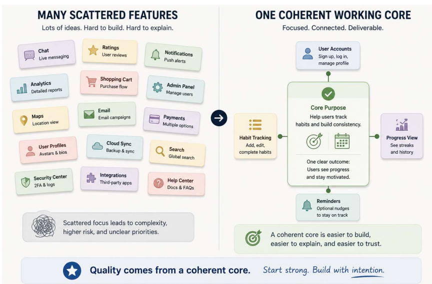


## Feature Count Is Not Project Quality

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## Scope Reduction Is A Professional Move

### Scope reduction asks:

- What is the core useful behavior?
- What can be removed?
- What can become a future feature?
- What must remain explainable?

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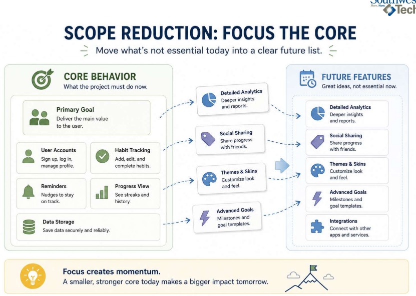


## Scope Reduction Is A Professional Move

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## Structure Should Match The Project

Possible structures:

- one clear script
- function-based program
- file/data utility
- API-style program
- simple class-based structure, if justified

*The structure should serve the project, not decorate it.*

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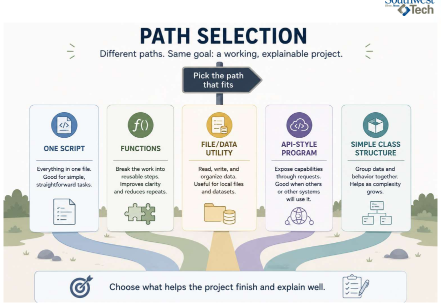


## Structure Should Match The Project Slide

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## Risks Should Be Named Before Coding

Risks are things that could make the project harder:

- live API instability
- unclear data format
- too many features
- logic not yet understood
- AI code the student cannot explain

*Naming risk early makes revision easier.*

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## Risks Should Be Named Before Coding

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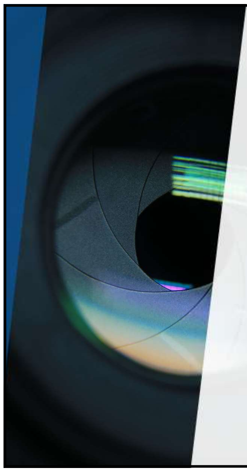

## Demo #1

"Scope Reduction Example"

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## What to Watch For

Watch the project shrink and improve.

- The goal is not to make the idea less interesting.
- The goal is to make the core project finishable.

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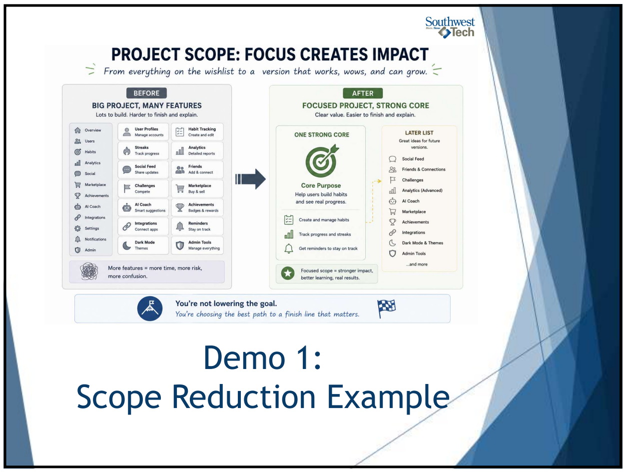
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PROJECT SCOPE: FOCUS CREATES IMPACT
From everything on the wishlist to a version that works, wows, and can grow.
BEFORE
BIG PROJECT, MANY FEATURES
Lots to build. Harder to finish and explain.
AFTER
FOCUSED PROJECT, STRONG CORE
Clear value. Easier to finish and explain.
ONE STRONG CORE
Core Purpose
LATER LIST
You're not lowering the goal.
You're choosing the best path to a finish line that matters.
Demo 1:
Scope Reduction Example

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Demo #2
"Structure Options Comparison"

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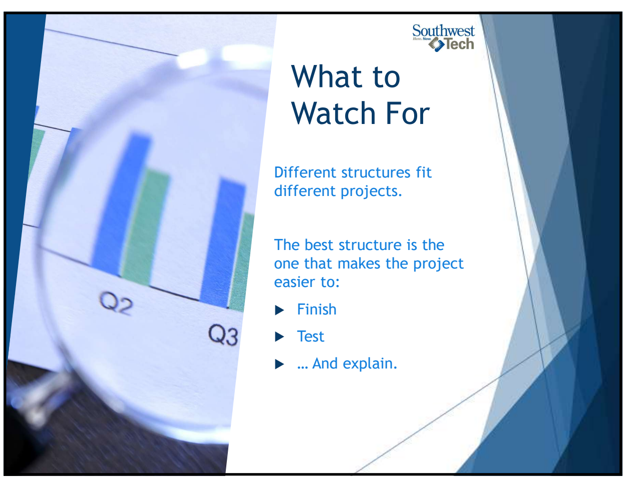
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What to Watch For
Different structures fit different projects.
The best structure is the one that makes the project easier to:
Finish
Test
... And explain.
Q2
Q3

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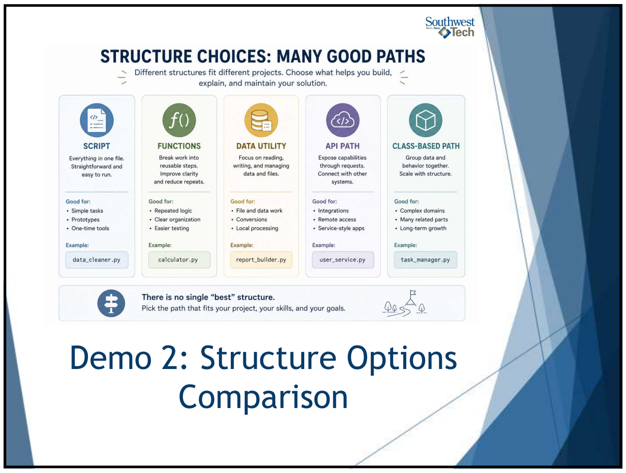
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STRUCTURE CHOICES: MANY GOOD PATHS
Different structures fit different projects. Choose what helps you build, explain, and maintain your solution.
SCRIPT
Everything in one file. Straightforward and easy to run.
Good for:
• Simple tasks
• Prototypes
• One-time tools
Example:
data_cleaner.py
FUNCTIONS
Break work into reusable steps. Improve clarity and reduce repeats.
Good for:
• Repeated logic
• Clear organization
• Easier testing
Example:
calculator.py
DATA UTILITY
Focus on reading, writing, and managing data and files.
Good for:
• File and data work
• Conversions
• Local processing
Example:
report_builder.py
API PATH
Expose capabilities through requests. Connect with other systems.
Good for:
• Integrations
• Remote access
• Service-style apps
Example:
user_service.py
CLASS-BASED PATH
Group data and behavior together. Scale with structure.
Good for:
• Complex domains
• Many related parts
• Long-term growth
Example:
task_manager.py
There is no single "best" structure.
Pick the path that fits your project, your skills, and your goals.
Demo 2: Structure Options Comparison

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Demo #3
"AI Boundary Example"

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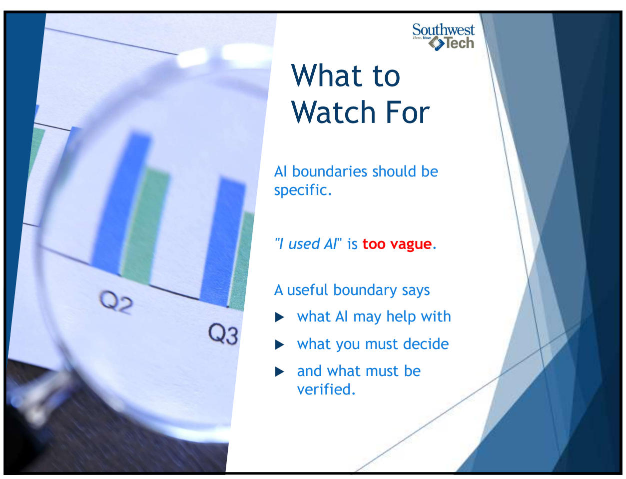
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What to Watch For
AI boundaries should be specific.
"I used AI" is too vague.
A useful boundary says
what AI may help with
what you must decide
and what must be verified.
Q2
Q3

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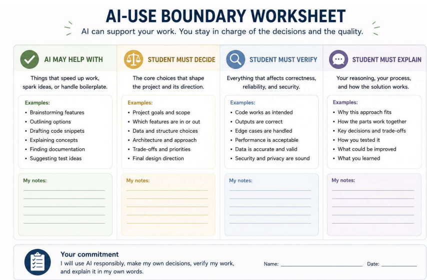


Demo 3:
AI Boundary Example

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Assignment #14

Today, Assignment 14 ***begins***.

Use your framing work to create a capstone proposal that defines

- Project name
- Purpose
- Features
- Inputs
- Outputs
- Structure
- Risks
- AI-use plan
- Realistic scope.

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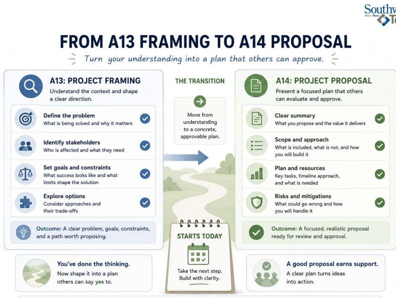


Assignment 14
(Starts Today)

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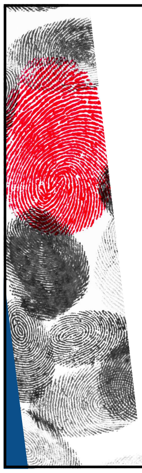

## Proposal Evidence

Your proposal should show:

- clear project purpose
- feasible feature list
- expected inputs and outputs
- likely structure
- risks or unknowns
- AI-use boundaries
- why the scope is realistic

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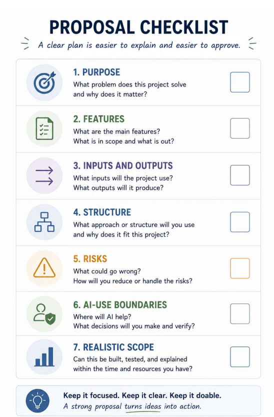


## Proposal Evidence

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## Approval May Require Revision

Approval is not just yes or no.

The instructor may approve, approve with limits, or request targeted revision so the project becomes more realistic.

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## Approval May Require Revision

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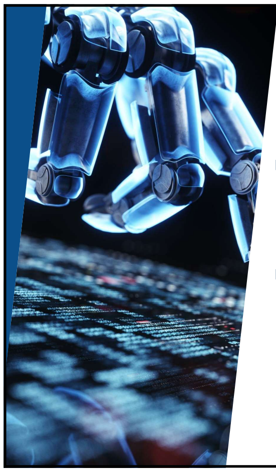

## AI Use

- AI can suggest API code that looks believable but does not match the actual response.
- You must inspect the response, run the code, and explain what human decision you made.

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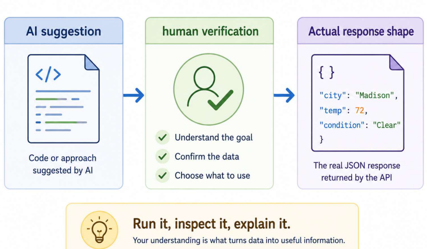


## AI-Assisted Code Must Be Verified

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## Closing Success Check

By the end of today, you should be able to say:

*"My proposal is smaller, clearer, more realistic, and easier to explain than my first idea."*

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## Success Check

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## Thank you!

## Have Fun!

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